

Table A. Leakage-controlled random 5-fold CV ablation results (mean R^2 across folds).

Model	Target	Mean CV R^2
M1 Base + independent	Km (mM)	-0.182
	kcat (s^{-1})	+0.408
	Vmax (nM s^{-1})	+0.085
M3 Physics-motivated + independent	Km (mM)	+0.345
	kcat (s^{-1})	+0.471
	Vmax (nM s^{-1})	+0.313
M4 Full (physics-motivated + chain)	Km (mM)	+0.345
	kcat (s^{-1})	+0.465
	Vmax (nM s^{-1})	+0.321
A0 Physics-motivated, no Arrhenius	Km (mM)	-0.204
	kcat (s^{-1})	+0.478
	Vmax (nM s^{-1})	+0.208
A1 Physics-motivated + activity label	Km (mM)	+0.594
	kcat (s^{-1})	+0.424
	Vmax (nM s^{-1})	+0.363
A2 Physics-motivated + Arrhenius	Km (mM)	+0.345
	kcat (s^{-1})	+0.465
	Vmax (nM s^{-1})	+0.321
A3 Physics-motivated + activity + Arrhenius	Km (mM)	+0.613
	kcat (s^{-1})	+0.409
	Vmax (nM s^{-1})	+0.350

Random CV quantifies predictive utility within the curated dataset and should not be interpreted as cross-publication validation. All models were evaluated on identical outer folds using fold-safe preprocessing; evaluation metrics were computed in log10 space on raw (unwinsorized) validation targets. M1/M3/M4 isolate the contributions of the engineered physics-motivated representation and RegressorChain; A0-A3 isolate activity-label and Arrhenius contributions.

Table B. Paired same-fold delta R^2 for the principal ablation comparisons (mean +/- SD across 5 folds).

Comparison	Target	Paired delta R^2 mean +/- SD
M3 - M1: engineered representation	Km (mM)	+0.527 +/- 0.389
	kcat (s^{-1})	+0.063 +/- 0.075
	Vmax (nM s^{-1})	+0.228 +/- 0.161
M4 - M3: RegressorChain	Km (mM)	+0.000 +/- 0.000
	kcat (s^{-1})	-0.006 +/- 0.032
	Vmax (nM s^{-1})	+0.008 +/- 0.058
A1 - A0: activity label	Km (mM)	+0.798 +/- 0.347
	kcat (s^{-1})	-0.054 +/- 0.090
	Vmax (nM s^{-1})	+0.155 +/- 0.180
A2 - A0: Arrhenius	Km (mM)	+0.549 +/- 0.393
	kcat (s^{-1})	-0.013 +/- 0.062
	Vmax (nM s^{-1})	+0.113 +/- 0.122
A3 - A1: Arrhenius beyond activity	Km (mM)	+0.019 +/- 0.023
	kcat (s^{-1})	-0.015 +/- 0.009
	Vmax (nM s^{-1})	-0.013 +/- 0.021

Positive values indicate improvement of the second condition over the first on the same outer fold. No qualitative effect-size labels are assigned. The largest gain for Km comes from the engineered representation; RegressorChain adds little; and the incremental Arrhenius contribution after explicit activity encoding is small.

Table C. Full-model performance under within-dataset and publication-aware validation.

Target	Random 5-fold CV R^2 mean +/- SD	Publication-grouped 5-fold CV R^2 mean +/- SD	Pooled grouped OOF R^2	OOF MAE log10	OOF within 2x
Km (mM)	+0.345 +/- 0.363	-0.164 +/- 0.643	+0.074	1.092	9.3%
kcat (s^{-1})	+0.465 +/- 0.106	-0.442 +/- 0.793	-0.014	2.550	9.3%
Vmax (nM s^{-1})	+0.321 +/- 0.116	-6.198 +/- 6.762	-0.803	1.650	25.6%

Publication-aware validation used GroupKFold with 23 reconstructed publication/source groups across all 86 records. The pooled OOF column combines one strictly out-of-publication prediction for every record. Random CV therefore represents within-dataset predictive utility, whereas grouped CV/OOF measures cross-publication transfer. The strongly lower grouped performance indicates substantial publication-level dependence and limited transfer to unseen studies.

Table D. Sensitivity analysis after excluding the unresolved 'unknown' activity class (n = 77).

Model	Target	CV R^2 mean +/- SD
A0 Physics-motivated, no Arrhenius	Km (mM)	-0.262 +/- 0.437
	kcat (s^{-1})	+0.366 +/- 0.117
	Vmax (nM s^{-1})	+0.257 +/- 0.172
A1 Physics-motivated + activity label	Km (mM)	+0.537 +/- 0.329
	kcat (s^{-1})	+0.253 +/- 0.226
	Vmax (nM s^{-1})	+0.222 +/- 0.185
A2 Physics-motivated + Arrhenius	Km (mM)	+0.164 +/- 0.287
	kcat (s^{-1})	+0.312 +/- 0.209
	Vmax (nM s^{-1})	+0.255 +/- 0.158
A3 Physics-motivated + activity + Arrhenius	Km (mM)	+0.551 +/- 0.310
	kcat (s^{-1})	+0.232 +/- 0.248
	Vmax (nM s^{-1})	+0.195 +/- 0.182

Excluding all nine unresolved activity annotations leaves the main qualitative conclusion unchanged: explicit activity encoding drives the largest Km improvement, while adding the Arrhenius descriptor on top of explicit activity changes mean CV R^2 only modestly (A3 vs A1).